MarketSage AI

Master Engineering Blueprint — **Part II**

Event Detection and Impact Scoring

The detection funnel, event taxonomy, empirically-recalibrated severity, novelty, the fake-news defence, event-study methodology and hierarchical Bayesian impact priors.

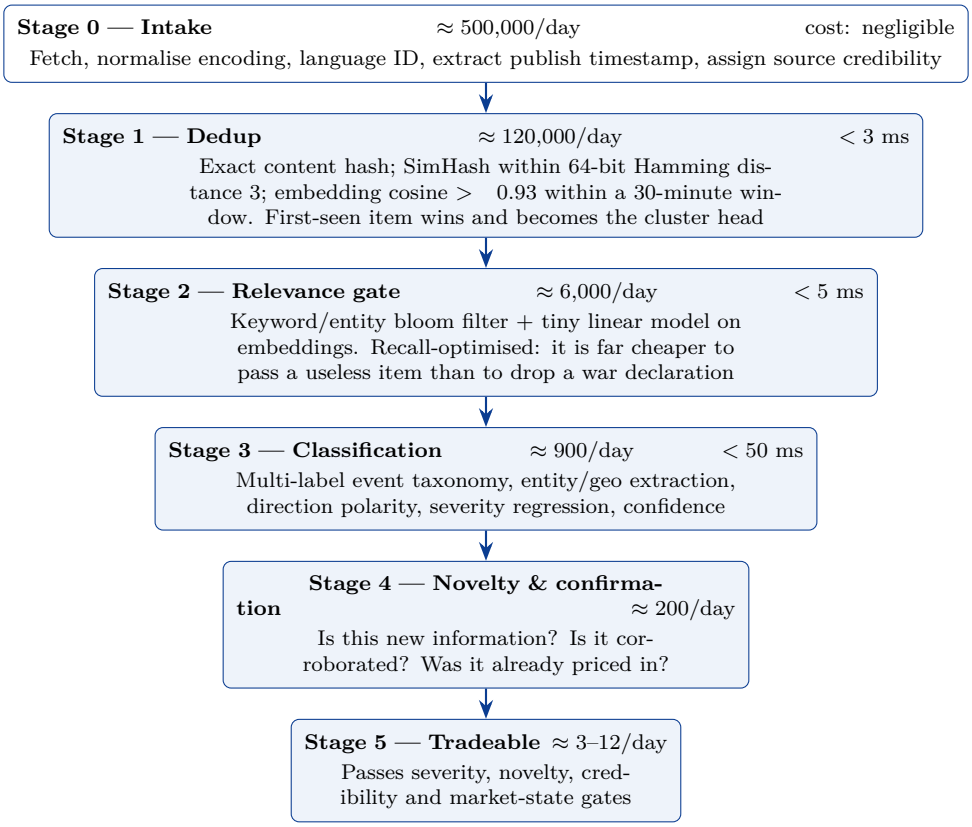
This is an extract of the MarketSage AI master plan, published separately for convenience. Chapter, section and equation numbering matches the full document, and cross-references point back into it. Read Chapter 3 (Reality Calibration) first — every design decision in this part follows from it.

CHAPTER 6

Event Detection Engine

6.1 The funnel

Roughly 500,000 items per day may enter the pipeline. Perhaps 40 are worth a model's attention and 3 are worth a trade. The engine is therefore a funnel with increasing cost per stage.



Asymmetric error policy

Stages 0–2 are tuned for **recall** (never lose a real event). Stages 3–5 are tuned for **precision** (never trade a non-event). Mixing these objectives in one model is the classic mistake.

6.2 Event taxonomy

The taxonomy is the backbone of the entire system: impact priors, features, model conditioning and post-trade attribution are all keyed on it. It must be designed once, carefully, and versioned.

Family	Classes (examples)	Typical severity	Primary Indian transmission channel
Monetary policy	rate decision, stance change, liquidity measure, unscheduled action, minutes/commentary	6–10	Bank Nifty directly; rate-sensitive sectors; currency

Family	Classes (examples)	Typical severity	Primary Indian transmission channel
Macro release	inflation, growth, employment, trade balance, industrial output, fiscal data	4–8	Broad index via surprise vs consensus
Geopolitical	military action, escalation rhetoric, territorial incident, sanctions, treaty, coup	3–10	Risk premium, crude, currency, gold; defence and oil-marketing sectors
Trade & tariffs	tariff announcement, export ban, supply-chain restriction	4–9	Export sectors (IT, pharma, chemicals, textiles)
Energy	production quota change, supply disruption, strategic reserve action, pipeline/refinery incident	4–9	Crude to inflation to rates; paint, aviation, oil marketing, tyres
Domestic policy	budget item, regulatory change, sector reform, tax change, subsidy	3–9	Sector-specific, occasionally index-wide
Corporate/market structure	index rebalancing, large block deal, big listing, margin/regulatory change	3–7	Flow-driven, highly mechanical, often the most predictable class
Flows & positioning	foreign flow reversal, large open-interest shift, expiry-day pinning	3–6	Mechanical, intraday, exploitable
Disaster & health	natural disaster, epidemic, major infrastructure failure	2–9	Insurance, infrastructure, supply chains
Political	election result, coalition change, major resignation, scandal	3–10	Broad risk premium; extremely high variance
Noise / non-event	opinion, analysis, repetition, speculation	0–2	**Explicit class** — the model must learn to say "nothing here"

Each classified event carries the following fields.

Field	Type	Meaning and how it is produced
`class`, `subclass`	enum	Multi-label transformer classifier.
`polarity`	$[-1, 1]$	Risk-off (-1) to risk-on ($+1$) for Indian equities specifically — *not* generic sentiment. A rate cut is negative sentiment about growth but positive polarity for equities.
`severity`	$[0, 10]$	Regression head trained on human-labelled anchors; described in Section 6.3.
`confidence`	$[0, 1]$	Classifier probability, temperature-calibrated.
`entities`	list	Countries, institutions, commodities, companies, sectors — linked to a canonical knowledge graph, not free text.

`geo`	list	Location plus a distance/exposure weight to India.
`novelty`	$[0, 1]$	Section 6.4.
`credibility`	$[0, 1]$	Source weight, boosted by corroboration.
`surprise`	$\mathbb{R}$	For scheduled events: $(\text{actual} - \text{consensus})/\sigma_{\text{consensus}}$. Null otherwise.
`expected`	bool	Was this event on the calendar? Scheduled and unscheduled events behave completely differently.

6.3 Severity: how to make it real and not arbitrary

A hand-written severity table will not survive contact with reality. Build severity in three passes:

1. **Anchor set.** Hand-label 400–600 historical headlines on a 0–10 scale using written rubrics with concrete anchors (10 = existential shock; 7 = major unexpected policy shift; 4 = meaningful but priced; 1 = commentary). Two labellers, measure inter-annotator agreement, resolve disagreements.
2. **Supervised head.** Train a regression head on the encoder to predict the anchor scores.
3. **Empirical recalibration.** This is the step everyone skips: regress the predicted severity against the *realised* absolute abnormal move of Nifty in the following 30 minutes across all historical events. Then remap severity so that it is monotonic in realised impact. Severity becomes an *empirically grounded* quantity rather than an opinion.

$$\text{severity}^{\text{cal}} = F^{-1}_{|r_{30}|}\Big(\hat{F}_{\text{sev}}\big(\text{severity}^{\text{raw}}\big)\Big)$$

where $\hat{F}_{\text{sev}}$ is the empirical CDF of raw severity and $F_{|r_{30}|}$ the empirical CDF of absolute 30-minute abnormal returns.

6.4 Novelty: is this actually new information?

The most expensive mistake is trading a re-report of something already priced. Novelty combines four signals:

$$\begin{aligned}\text{novelty} \;=\; & w_1 \underbrace{\big(1 - \max_j \cos(e, e_j)\big)}_{\text{semantic distance to last 6 h}} + w_2 \underbrace{\mathbb{1}[\text{first mention in cluster}]}_{\text{primacy}} \\ & + w_3 \underbrace{\big(1 - \rho_{\text{cal}}\big)}_{\text{not on calendar}} + w_4 \underbrace{\big(1 - |\hat{r}_{\text{pre}}|/\sigma_5\big)}_{\text{market has not already moved}}\end{aligned}$$

The fourth term is the most underrated: if the index has already moved two standard deviations in the five minutes *before* our detection, the information is in the price and the opportunity has passed. In that case the system should consider a *fade* strategy or, more usually, no trade at all.

6.5 Confirmation and the fake-news defence

Rule	**Condition**	**Consequence**
Single low-credibility source	$c < 0.5$, no corroboration	Hot-lane alert only. No directional signal. Volatility-widening only.

Corroboration window	Second independent source with $c \geq 0.75$ within 120 s	Credibility upgraded; directional signal permitted.
Official confirmation	Primary source (regulator, central bank, exchange, official channel)	Full weight immediately.
Contradiction	An equally credible source contradicts within 5 min	Emit `RETRACTION`; immediately close any position opened on that event and mark the event poisoned.
Extreme severity gate	severity ≥ 9	Requires **two** independent sources with $c \geq 0.8$ regardless of speed. The most extreme headlines are the most likely to be fake.
Market disagreement	Signal says down, but index, futures and currency all move up within 30 s	Suppress. The market is a better classifier than we are.

The AP-hack test

Before going live, run this test: inject a synthetic, highly credible-looking but false catastrophic headline from a single medium-credibility source into a replay. **The system must not produce a directional trade.** If it does, the confirmation logic is wrong. Repeat this test in CI forever.

6.6 Entity-to-instrument routing

An event mentioning "crude supply disruption" must mechanically become an exposure vector. Maintain a knowledge graph:

Listing 6.1: Exposure routing (conceptual)

```
# knowledge graph edge: (entity) --affects--> (factor) --loads_on--> (instrument)
EXPOSURE = {
    "crude_oil": {"NIFTY": -0.25, "BANKNIFTY": -0.15,
                  "sector:OIL_MKTG": -0.80, "sector:AVIATION": -0.70,
                  "sector:PAINTS": -0.55, "sector:UPSTREAM": +0.60},
    "usd_inr_up": {"NIFTY": -0.20, "sector:IT": +0.65, "sector:PHARMA": +0.40},
    "rbi_rate_cut": {"BANKNIFTY": +0.70, "NIFTY": +0.45,
                     "sector:REALTY": +0.75, "sector:AUTO": +0.55},
}
# Beta values are ESTIMATED FROM HISTORY (rolling regression), never hand-typed
# in production. The dict above is only the bootstrap prior for cold start.
```

Rule

Hand-written coefficients are acceptable only as a cold-start prior. Every coefficient must be replaced by a rolling empirical estimate with a confidence interval as soon as sufficient history exists. The system must be able to answer "how do you know?" with a number and a sample size.

CHAPTER 7

Impact Scoring Engine

7.1 What impact scoring actually is

The original design computes "Nifty will drop 1.5%". That is a point estimate with no error bar, and it is the root cause of over-sizing. The correct object is a **conditional predictive distribution**:

$$P(r_{t\to t+h} \mid \text{event class } c,\ \text{severity } s,\ \text{surprise } z,\ \text{market state } m)$$

produced primarily from historical evidence, and used as a *prior* that the ML models in Chapter 9 then update with live microstructure.

7.2 Building the impact library: event study methodology

For every historical event of class c:

1. Define the event time t_0 precisely (publish timestamp, exchange time).
2. Compute the **abnormal return** over horizon h: subtract the expected return given the market's normal behaviour at that time of day and volatility regime,

$$AR_h = r_{t_0\to t_0+h} - \hat{\mathbb{E}}[r_h \mid \text{time-of-day}, \sigma_{\text{regime}}, \text{day-type}]\,.$$

   Time-of-day normalisation matters enormously in Indian intraday data: the volatility profile is strongly U-shaped, so a raw 0.3% move at 09:20 is ordinary while the same move at 13:30 is extreme.
3. Compute a matched **control distribution**: sample non-event windows with the same time-of-day and volatility regime.
4. Record: AR_h for $h \in \{1, 5, 15, 30, 60\}$ minutes, realised volatility ratio, implied volatility change, maximum favourable and adverse excursion, volume ratio, and time-to-peak.
5. Test significance against the control distribution with a bootstrap, not a normal approximation (intraday returns are fat-tailed).

Stored statistic	**Why it matters for trading**
Median AR_h and quantiles	The realistic expectation and its spread.
P(sign correct)	Whether a directional trade is even sensible.
Realised-vol ratio $\sigma_{\text{post}}/\sigma_{\text{pre}}$	Whether to buy volatility.
$\Delta\sigma_{\text{IV}}$ distribution	**The vega decision.** Determines whether options get more or less expensive.
Max favourable / adverse excursion	Where to put the target and the stop, from evidence rather than a round number.
Time-to-peak	The correct holding period; also determines the horizon to trade.
Half-life of the effect	Whether the move persists or mean-reverts.
Sample size n	The honesty check.

7.3 Hierarchical Bayesian shrinkage for rare events

Most interesting events are rare. With $n = 4$ historical instances, the empirical mean is noise. Use a hierarchical model that shrinks class-level estimates toward their family-level estimate:

$$\mu_c \sim \mathcal{N}(\mu_{\text{family}}, \tau^2), \qquad AR_{c,i} \sim \mathcal{N}(\mu_c \cdot g(s_i), \sigma_c^2)$$

with posterior mean approximately

$$\hat{\mu}_c = \frac{n_c/\sigma_c^2}{n_c/\sigma_c^2 + 1/\tau^2}\, \bar{AR}_c \;+\; \frac{1/\tau^2}{n_c/\sigma_c^2 + 1/\tau^2}\, \mu_{\text{family}}.$$

Practical consequence

With $n_c = 3$, the estimate is dominated by the family prior — which is exactly correct behaviour. The engine reports the posterior *and* the sample size, and the risk gate reduces position size when the posterior variance is high. Refuse to emit a class-specific impact when $n_c < 8$ and no family prior exists; fall back to "unknown event, volatility-only response".

7.4 The live scoring formula

$$\hat{r}_h = \underbrace{\hat{\mu}_{c,h}}_{\text{Bayesian prior}} \times \underbrace{g(s)}_{\text{severity}} \times \underbrace{\nu}_{\text{novelty}} \times \underbrace{c_{\text{cred}}}_{\text{credibility}} \times \underbrace{\kappa(m)}_{\text{regime}} \times \underbrace{\lambda(\tau)}_{\text{time-of-day}} \times \underbrace{(1-\pi)}_{\text{already-priced discount}}$$

Term	Form	Notes
$g(s)$	$(s/5)^\gamma$, $\gamma \approx 1.4$ fitted	Impact is convex in severity: a 9 is much more than twice a 4.5.
ν	novelty score	Old news gets multiplied toward zero.
c_{cred}	credibility	Unconfirmed rumour is heavily discounted.
$\kappa(m)$	regime multiplier	Same event has larger impact in a high-volatility, low-liquidity regime. Fitted per regime cluster.
$\lambda(\tau)$	time-of-day multiplier	U-shaped; also handles the special cases of the opening 15 minutes, the pre-close and expiry-day afternoons.
π	$\min(1, \lvert r_{\text{pre}}\rvert / \lvert\hat{\mu}_{c,h}\rvert)$	If the market has already made the expected move before we act, the remaining opportunity is gone. **This term is what prevents chasing.**

The output is not a scalar but a triple: ($\hat{r}_h$, quantiles, n_{eff}) for each horizon h, plus a separate volatility forecast ($\hat{\sigma}_{\text{post}}/\sigma_{\text{pre}}$, $\widehat{\Delta\sigma_{IV}}$).

7.5 Cross-asset propagation

When an event occurs outside Indian market hours, or affects a foreign asset first, the impact reaches Nifty through a measurable chain. Estimate the chain empirically with lagged regressions and store the transfer coefficients:

Channel	Typical lag	Modelling note

GIFT/SGX Nifty to Nifty spot	Near-instant during overlap; sets the gap when Indian market is shut	The single most useful overnight variable.
US equity futures to Indian open	Overnight	Estimate a gap model: $r_{\text{gap}} = \alpha + \beta\, r_{\text{US futures}} + \ldots$
Crude to Nifty	Minutes to hours	Asymmetric: sharp spikes transmit faster than slow drifts.
USD/INR to Nifty	Minutes	Bidirectional causality; use a vector autoregression, not a simple regression.
US Treasury yields to Bank Nifty	Hours	Slower; more relevant to the next session.
Asian indices to Nifty intraday	Minutes	Useful during the 09:15–12:00 window.

The overnight opportunity

Roughly half of the total price change in Indian indices arrives as overnight gaps. Most retail systems ignore this because they only look at intraday candles. A well-calibrated **gap forecaster** driven by overnight global events is arguably an easier and more profitable first product than intraday prediction — and it needs no low-latency infrastructure at all. Consider shipping it first (see Phase 4 in the roadmap).